Learning Noisy Label Pseudo-Refinements Without Meta-Gradient

Zhanhui Lin [1] **Sanping Zhou** [1]

Abstract

Efficient use of noisy data for supervised learning has become a crucial research area, due to the high cost of acquiring large-scale, accurately labeled datasets for modern deep learning area. Previous studies have demonstrated significant improvements in learning outcomes when a small amount of clean data is incorporated. In this context, meta-learning-based methods have proven particularly successful, especially label correction techniques. However, these methods are computationally intensive, as they involve calculating high-order derivatives at each training step, which significantly slows down the training process. In this study, we introduce a novel model-agnostic methodology for leveraging limited quantities of clean data, significant depart from traditional techniques that often rely on estimating a transition matrix or employing meta-gradient adjustments. Our approach focuses on directly utilizing the limited clean data to guide the refinement of noisy labels with various stages of model posteriors with decoupled bi-level optimizations in a self-training framework. Our method achieves 79.62% accuracy on the Clothing1M dataset without fine-tuning on the clean dataset, with apparently lower overheads than existing leading methods, demonstrating the effeteness of our method in the real-world application. The code for our experiments is accessible at `https://anonymous.4open.science/r/noisy_refinements`.

1. Introduction

The success of deep neural networks in various domains highlights the importance of large-scale data sources in achieving high-performing models. However, obtaining a completely clean dataset for real-world production can often be prohibitively expensive. When direct supervised learning is applied to noisy datasets, the neural network tends to overfit the noise present in the data. This phenomenon leads to a degradation in model performance and generalization (Zhang et al., 2021; Shwartz-Ziv & Tishby, 2017).

Multiple methods have been developed to effectively use noisy training data for robust learning. Pseudo-labeling for label correction techniques are increasingly popular in this field (Chen et al., 2023; Li et al., 2020). These methods make use of pseudo-labels derived from network predictions to utilize the information within the data based on semi-supervised learning principles, which are particularly effective in dealing with instance-dependent noise in real-world scenarios. Nevertheless, the manual tuning of weights has presented a significant challenge, often resulting in less-than-ideal performance and a intricate optimization process. In response, the meta-learning framework has been employed to address this issue, as described in (Karim et al., 2022; Wu et al., 2021), which automate the weight allocation process. These methods dynamically adjust the balance between the observed noisy labels and the model predictions. They utilize a small clean dataset to calculate meta-gradients, which update the model architecture or other hyper-parameters to make them robust to the noisy data.

In fact, while it is challenging to obtain an entirely clean dataset, it is still feasible and practical to accurately label a small part of the data. Early research has shown significant performance improvements when a small amount of reliable data is used to assist training (Li et al., 2017; Hendrycks et al., 2018). Over time, meta-learning-based methods have often been more advantageous on final performance than other methods. Despite the effectiveness of the meta-learning-based methods, they have limitations. Their reliance on clean data often involves implicit optimization, which can restrict the interpretability of the model and lead to over-reliance on previously miscorrected labels (Kye et al., 2022). Moreover, a significant drawback is their requirement for multiple virtual updates within the inner loop. This leads to substantial computational overhead (Wang et al., 2020; Zhang et al., 2019) and larger memory usage, severely limiting their efficiency and feasibility for training on large-scale datasets.

[1]Institute of Artificial Intelligence and Robotics, Xi'an Jiaotong University, Xi'an 710049, China. Correspondence to: , Sanping Zhou <spzhou@xjtu.edu.cn>.

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To address the aforementioned challenges, we introduce an innovative learnable pseudo-refinement technique founded on a bi-level optimization framework, distinct from traditional meta-learning approaches. Our approach involves training the classifier utilizing refined pseudo-labels within the inner loop, while the outer loop learns the optimal coefficient and residual modeling function using limited clean data to facilitate the pseudo-relabeling strategy, with a bridge connecting the learned refined labels from the noisy distribution to the clean ones. Our method leverages the ensemble of predictions across various training stages, capturing a broader spectrum of information to correct label inaccuracies. It systematically enhances auxiliary posteriors over different training iterations, refining labels and subsequently retraining the model on these improved pseudo-labels. The separation of objectives and the streamlined structure allows for more efficient optimization and flexibility and reduces the memory usage and computational overhead of multiple back-propagations.

We validate our approach with comprehensive experiments on the CIFAR and Clothing1M datasets, including examining the impacts of auxiliary clean data scale and corruption leakage into the clean data. Our results demonstrate that our method not only significantly accelerates training but also achieves notable performance improvements. This enhancement presents a compelling solution for tackling large-scale noisy label learning, offering an efficient, robust, and data-driven approach that effectively leverages limited clean data to rectify and refine noisy labels.

2. Related Work

A breadth of studies have explored strategies to mitigate the impact of noisy labels, employing diverse techniques and perspectives. Theoretically analyzing the impact of label noise has provided insights into when and why noise hurts performance (Oyen et al., 2022; Adebayo et al., 2022; Bar et al., 2022). Many works have developed noise-robust loss functions and regularization to prevent overfitting (Wei et al., 2021; Zhang & Sabuncu, 2018; Liu et al., 2022). Reweighting (Jiang et al., 2018; Liu & Tao, 2015) and sample selection (Li et al., 2020; Xia et al., 2021) techniques aim to reduce the influence of likely mislabeled examples. Transition matrix estimation methods model the noise process for learning with consistencies (Xia et al., 2019; Yao et al., 2020; Yang et al., 2021). Some studies have focused on enhancing optimization techniques to alleviate the impact of noise (Liu et al., 2020; Gorbunov et al., 2020). Semi-supervised learning leverages unlabeled or noisy data to combat label noise (Pham et al., 2021; Schmarje et al., 2022; Han et al., 2019). More recent efforts (Cheng et al., 2020; Chen et al., 2021; Ma et al., 2022) tackle instance-dependent label noise, which better matches real-world settings.

3. Preliminary

We address the problem of c-class classification in this study. The feature space is represented as $\mathcal{X} \subseteq \mathbb{R}^d$, and the label space is denoted by $\mathcal{Y} \subseteq \mathbb{R}^c$, where c signifies the number of classes. We define the noisy label dataset as $\widetilde{D} = \{(x_i, \widetilde{y}_i)\}_{i=1}^N$, where each $\widetilde{y}_i$ is an observed noisy label. This dataset is assumed to be sampled from a noisy label distribution $p_{x,\widetilde{y}}$. Correspondingly, we have a latent true dataset $D = \{(x_i, y_i)\}_{i=1}^N$, which contains the true labels for each feature vector x_i. In $\widetilde{D}$, each sample $(x_i, \widetilde{y}_i) \in (\mathcal{X}, \mathcal{Y})$ is associated with a possibly noisy label $\widetilde{y}_i$, while in D, the same feature x_i is paired with its true label y_i. Label noise can generally be categorized as either instance-independent or instance-dependent, with further details and explanations provided in Appendix A. We also incorporate a subset of trusted data, known as the clean dataset $D^{(m)} = \{({x^{(m)}}_i, {y^{(m)}}_i)\}_{i=1}^M$, sourced from the clean label distribution $p_{x,y}$. Here, in the clean dataset D, each sample (x, y) has its true label y. We assume that the sizes of the noisy training set and the clean set satisfy $M \ll N$ which satisfies real-world settings.

Let $f(x)$ be a neural network that predicts the classification. We assume that the final output layer is a softmax layer, which approximates the class-conditional probabilities $p(y|x)$. Our primary objective is to train an optimal classifier using the noisy dataset to minimize the expected risk on unseen test data.

4. Method

4.1. Existing Methods that Learn From Clean Data

Current methods in training with noisy labels predominantly utilize clean data, mostly falling into two main categories: transition-matrix-based methods, which model transitions from clean to noisy labels, and meta-learning-based approaches, increasingly popular for adaptive hyper-parameter optimization in noise modeling.

Meta-learning-based Methods: These methods use a clean dataset to adjust the learning process of a model through hyper-parameter optimization (Liu & Tao, 2015; Zheng et al., 2021). While they effectively enhance label accuracy by leveraging clean data, their computational intensity and memory demands often limit their scalability, especially for large datasets.

Transition-matrix-based Correction Methods: Transition matrix approaches correct noisy labels by bridging the gap between the class posterior probabilities of noisy and clean data (Pene et al., 2021; Yang et al., 2021). They primarily address instance-independent noise but can struggle with instance-dependent noise and suffer from performance issues due to the variance in one-hot noisy labels .

For a detailed discussion and mathematical formulation of these methods, see Appendix B.

4.2. The Proposed Learnable Pseudo-Refinements

To tackle these problems, we introduce our learnable pseudo-refinement method, a simple but novel approach use of the limited clean data, designed to refine the noisy labels in the noisy training dataset.

4.2.1. Principle and Formations

We assert that the refined label can always be decomposed into a convex combination of the original noisy label and a residual term, expressed as:

$$\hat{y} = \alpha \cdot \widetilde{y} + (1-\alpha) \cdot \overline{y}, \tag{1}$$

where $\hat{y} \in \mathcal{Y}$ denotes the refined pseudo-label, which is a refined version of the original label $\widetilde{y}$. The parameter α, ranging from 0 to 1, controls the contribution of the original label to the pseudo-label. The term $\overline{y} \in \mathcal{Y}$ represents the residual correction behavior, a crucial element that adjusts the label towards its true value.

Definition 4.1. Define the expected risk under the clean label distribution as

$$R(y') = \mathbb{E}_{(x,y)\sim p_{x,y}} \mathcal{L}(y', y), \tag{2}$$

where y denotes the true label, and y' the given arbitrary label. The empirical risk over a sample set of size M is defined as

$$R_M(y') = \frac{1}{M} \sum_{i=1}^{M} \mathcal{L}(y'_i, y_i), \tag{3}$$

where y_i represents the true label and y'_i the arbitrary label for the k-th sample.

Proposition 4.2. *The risk associated with the pseudo-label $\hat{y}$ is never greater than the risk associated with the original noisy label:*

$$\min_{\alpha,\overline{y}} R(\hat{y}) \leq R(\widetilde{y}). \tag{4}$$

Proof. When $\alpha = 1$, Equation 1 reduces to $\hat{y} = \widetilde{y}$, making $R(\hat{y}) = R(\widetilde{y})$. For $\alpha \in [0,1]$ and varying $\overline{y}$, minimizing $R(\hat{y})$ yields an infimum over the feasible set of α and $\overline{y}$. Since $R(\widetilde{y})$ represents a specific case of $R(\hat{y})$ at $\alpha = 1$, it follows that the infimum of $R(\hat{y})$ is at most $R(\widetilde{y})$. □

Lemma 4.3. *For any given $\overline{y}$, a configuration of α and $\hat{y}$ exists, such that $R(\hat{y}) \leq R(\widetilde{y})$, with at least one configuration where the inequality is strict.*

Lemma 4.3 reveals a counter-intuitive yet fascinating result: the amalgamation of a *highly noisy* data distribution can, in fact, contribute to the purification of noisy labels. In subsequent sections, we introduce a function g dedicated to computing the residual term. Theoretically, the inputs to g can be varied, encompassing diverse forms of prior knowledge and adapting to different data conditions. The definition can be expressed as

$$\overline{y} = g(.). \tag{5}$$

The key of our method involves harnessing the intrinsic information present in the noisy labels, while concurrently rectifying their inaccuracies by employing a meticulously calibrated residual term in conjunction with the weighting coefficient. We formulate the following bi-level minimization problem to jointly correct the noisy labels and learn the classifier parameters:

$$\alpha^*, g^* = \underset{\alpha\in[0,1], g\in\mathcal{G}}{\arg\min} \frac{1}{M} \sum_{i=1}^{M} \mathcal{L}_M(\hat{y}_i^{(m)}(\alpha, g), y_i^{(m)}), \tag{6}$$

$$f^*_{\alpha,g} = \underset{f\in\mathcal{F}}{\arg\min} \frac{1}{N} \sum_{i=1}^{N} \ell(f(x_i), \hat{y}_i(\alpha, g)), \tag{7}$$

where $\mathcal{L}_M$ is loss function imposed on clean data. A remained challenge arises from Equation 6, wherein the corresponding noisy label $\widetilde{y}^{(m)}$ for $\hat{y}^{(m)}$ is not directly available. Given that $\widetilde{y} \sim p(\widetilde{y}|x)$, we define $h(x)$ to represent this relationship, i.e., $h(x) = p(\widetilde{y}|x)$. Subsequently, we determine the pseudo-label for the clean dataset through

$$\hat{y}^{(m)} = \alpha \cdot h(x^{(m)}) + (1-\alpha) \cdot g(.). \tag{8}$$

Our approach effectively synchronizes the training process with the clean data distribution. We assume that the dimensions of the label space c and the feature space d satisfy the condition $c \ll d$, and the sample counts in both the noisy training set and the clean set conform to $M \ll N$, facilitating optimization to be more efficient in the outer loop than inner loop.

The proposed learnable pseudo-relabeling technique offers several advantages compared to previous methods that made use of the clean data introduced in Section 4.1, outlined as follows:

- **Optimization Efficiency:** The separation of the inner and outer loops in our approach accelerates the optimization process, avoiding step-by-step optimization in meta-learning framework and significantly reducing memory requirements and computational complexity. This separation also enables the use of non-gradient-based optimization methods, enhancing the overall efficiency of the model training.
- **Flexible Adjustments:** In contrast to traditional meta-learning methods where the forms of hyper-parameters

Θ are typically fixed, our pseudo-relabeling approach allows for dynamic adjustment of parameters within the function g. This flexibility provides a more adaptable framework, capable of handling various data conditions throughout the training process. This approach not only facilitates more effective management of different data types but also fosters a deeper understanding of refinement behaviors.

- **Adaptability and Robustness:** Unlike transition-matrix-based methods, our approach does not rely on the assumption of instance-independent noise. Moreover, our approach may also adapt to elegantly handle continuous or multi-label problems which would be hard to modeled with transitions, showcasing its versatility and robustness.

4.2.2. Modeling Residuals with Posteriors

We employ innovative strategies to leverage a comprehensive spectrum of information through self-training predictions at various stages of the model's development, avoiding the extra cost in teacher-student framework (Jiang et al., 2018; Taraday & Baskin, 2023).

Throughout the training process, we define $\widetilde{y}_t$ and $\widetilde{y}_t^{(m)}$ as the predictions corresponding to the model f_t applied on instance $(x, \widetilde{y})$ in the noisy dataset $\widetilde{D}$ and instance $(x^{(m)}, y^{(m)})$ in the small-scale clean dataset $D^{(m)}$, respectively. Here, f_t represents the model at distinct phases of training. Our methods go beyond the conventional use of only the latest pseudo-labels in a bootstrapping style. By aggregating these predictions, we harness diverse information sources to achieve a more stable and accurate label correction. We explore several distinct techniques, ensuring a more nuanced and comprehensive understanding of the evolving data landscape during model training. We note that for all the functions g introduced below, $\widetilde{y}_t^{(m)}$ is processed equivalently to $\widetilde{y}_t$, while $\widetilde{y}^{(m)}$ is approximated using noisy posterior $h(x^{(m)})$.

Label Mixed up With/Without Temperatures: One natural technique strategically blends pseudo-labels obtained with a set of linear coefficients array λ, enabling a balanced and controlled integration of labels across various model stages, expressed as:

$$g_\lambda(\widetilde{y}, \{\widetilde{y}_t\}_{t=1}^n) = \sum_{t=1}^n \lambda_t \widetilde{y}_t. \tag{9}$$

While some studies (Reed et al., 2014; Karim et al., 2022) rely on the $\arg\max$ operation to derive pseudo-labels $\widetilde{y}$, we contend that this approach may lead to a loss of valuable information. To address this, we draw inspiration from early research (Li et al., 2017) and introduce a learnable temperature parameter array τ, to modulate the model outputs. We posit that the intermediate logits, z_t, produced by f_t before the softmax-induced prediction $\widetilde{y}_t$, retain richer information, especially than all the elements of τ are initially set to 1. To integrate this concept, we modify Equation 9 to include a learnable temperature, formulated as follows:

$$g_{\lambda,\tau}(\widetilde{y}, \{z_t\}_{t=1}^n) = \sum_{t=1}^n \lambda_t s_t, \tag{10}$$

where s_t is the soft label under τ_t, the j-th element in s_t is

$$s_t[j] = \frac{\exp\left(z_t[j]/\tau_t\right)}{\sum_j \exp\left(z_t[j]/\tau_t\right)}. \tag{11}$$

We test that integrating this temperature parameter can enhance our formulation to consider the variability in information content across different model outputs, potentially offsetting biases and variances inherent in individual model stages. This methodology can provides a more robust and information-rich framework for label correction and model refinement.

Neural Network Correction: The second technique leverages a small simple neural network parameterized by θ whose last activation is softmax and output dimension is c, symbolized as $\mathcal{NN}$, to intelligently amalgamate pseudo-labels from different training phases, formulated as:

$$g_\theta(\widetilde{y}, \{z_t\}_{t=1}^n) = \mathcal{NN}\left(\text{concat}\left(\widetilde{y}, \{z_t\}_{t=1}^n\right)\right). \tag{12}$$

This method employs the neural network's inherent capacity to provide a sophisticated mechanism for label refinement.

4.2.3. Overall Algorithm

Our methodology is centered on iteratively expanding auxiliary useful information, which are stored in two buffer pools $\mathcal{Z}$ and $\mathcal{Z}^{(m)}$, and harnessing them to refine the noisy labels with the clean dataset. This process culminates in a streamlined algorithmic framework after training directly on the noisy dataset to obtain the noisy posterior function h in the first phase, as outlined in Algorithm 1.

In Section 4.2.2, the formulations of function g exhibit distinct parameterizations, necessitating a choice based on the size of the clean dataset at hand. Moreover, to counter potential overfitting due to the limited volume of clean data, a strategic division into training and validation subsets is essential. The optimal configuration for α and g is determined by minimizing the validation error on this subset, which then guides label correction during the main training phase. Additionally, the clean data serves as a reward metric, allowing for experimentation with different forms of g in each outer loop, and the most effective variant is then chosen based on its ability to minimize the error on the clean dataset.

Algorithm 1 Learning Label-Refinement Streamline

Input: Dataset $\widetilde{D}$ and $D^{(m)}$, Pool $\mathcal{Z}$, $\mathcal{Z}^{(m)}$.
Output: Prediction Model f

Initialize model h and train h on $\widetilde{D}$
Infer $\{(x_i^{(m)}, h(x_i^{(m)}))\}_{i=1}^{M}$ on $D^{(m)}$, update $\mathcal{Z}^{(m)}$.
Initialize model f_1, $t = 1$.
Set $\alpha = 1$, $g =$ null.
while not convergence **do**
  for each training epoch **do**
    Train f_t with Eq. 7 using $\mathcal{Z}$.
  end for
  Infer $\{(x_i, z_{t,i}))\}_{i=1}^{N}$ on $\widetilde{D}$, update $\mathcal{Z}$.
  Infer $\{(x_i^{(m)}, z_{t,i}^{(m)}))\}_{i=1}^{M}$ on $D^{(m)}$, update $\mathcal{Z}^{(m)}$.
  Reset α, g, Obtain α^*, g^* with Eq. 6 using $\mathcal{Z}^{(m)}$.
  $t+ = 1$.
end while

We emphasize that algorithm does *NOT* require multiple classifier to infer their predictions at each training iteration. Instead, it only infer and records the logits once for subsequent outer loop computations and analysis. Although our method incorporates a two-stage training process, it still expedites the overall training compared to the multiple back-propagations typically in meta-learning approaches. Furthermore, our method offers practical flexibility, such as the potential for engineering trade-offs by saving checkpoints and re-launch, or by slightly modifying the model architecture, as suggested in (Kye et al., 2022) to more effectively estimate a stable noisy posterior while concurrently training the target classifier to completely omit the first noisy posterior training phase.

5. Experiments

5.1. Experiments on CIFARs

We first evaluate our method on CIFAR-10 and CIFAR-100 (Krizhevsky et al., 2009) which is widely used for machine learning research. Both datasets contain 50,000 training and 10,000 test images. We randomly select 5,000 images from the training set as the clean dataset.

Noise settings: We simulate two types of **Instance-independent** noise. **Symmetric**: Generated by flipping the labels of a given proportion of training samples to all the other class labels uniformly. Each instance with a true label y is corrupted to all possible classes with probability $\frac{\eta}{c}$. **Asymmetric**: Generated by flipping the labels of a given proportion to one or few particular classes. For CIFAR-10, we disturb the label to its similar class with probability η follow (Yao et al., 2019). For CIFAR-100, we disturb the label to other classes in its super-class as described in(Hendrycks et al., 2018). Noise transition matrices are first generated and labels are randomly corrupted according to the matrix for instance-independent label noise. Additionally, we evaluate **Instance-dependent** noise as described in (Zhu et al., 2021) on both datasets. We vary η in the range of $[0, 1]$ to simulate different noise levels for both types.

Baselines: We compare our methods with a series of advanced methods in noisy label learning that use clean data. The compared methods include: *Transition-Matrix-Based Methods*: **GLC.** (Hendrycks et al., 2018) estimates the noise transition matrix with a trained noisy posterior. **FasTEN.** (Kye et al., 2022) introduce a two-head architecture to efficiently estimate the label transition matrix every iteration within a single back-propagation. *Meta-Learning-Based Methods*: **MW-net.** (Shu et al., 2019) uses an MLP net and meta-learns the weighting function. **L2B.** (Karim et al., 2022) uses a bootstrapping method with meta-learned weighting coefficient. **EMLC.** (Taraday & Baskin, 2023) proposes a teacher-and-student architecture and meta-learns the taught soft-label. Notably, for FasTEN and GLC, we modified their standard training procedure by removing the loss component corresponding to the clean instances, in order to maintain experiment consistency.

Implementations: In our experiments on CIFAR-10 and CIFAR-100, we employed ResNet-34 as the backbone architecture. To ensure a fair comparison across different methods, each was trained from scratch under identical conditions, including the use of the same data augmentation techniques. We train all the models with SGD with batch size 128 for 100 epochs with LR-step scheduler at 60 and 80 epochs. Other detailed specifications of our experimental settings are provided in Appendix C. To accurately assess the models' capability in handling label noise, we report the final accuracies for each experiment without the use of early stopping or similar techniques.

For our method's specific settings, we utilize SGD to optimize the refinement parameters on the training subset, which constitutes 80% of the clean dataset. The remaining 20% is reserved for the validation subset, from which we select the optimal parameters based on the best performance. In mixed-up strategies, we optimize the weighting coefficients array under softmax for α and λ_t, including temperature terms τ_t when applicable. The neural network correction employs a 2-layer MLP, with hidden units of 128 and a ReLU activation in between, followed by a softmax output layer, facilitating dimensionality transformation from $c \times (n + 1)$ to c. We periodically update $\mathcal{Z}$ and $\mathcal{Z}^{(m)}$ every 10 epochs, after loading the checkpoint from the first training phase at epoch 40. We claim that the primary purpose of loading the checkpoint is to save time, with minimal impact of no warm-up behaviors. In our formulation, as delineated in Equation 6, an automatic weighting is applied to diminish the influence of muddled posteriors.

Table 1. Test accuracy (%) of all competing methods on CIFAR-10 and CIFAR-100 under symmetric and asymmetric noise with different noise levels. The best results are highlighted in bold.

DATASETS	METHOD	SYMMETRIC			ASYMMETRIC			INSTANCE-DEPENDENT		
		η			η			η		
		0.2	**0.4**	**0.6**	**0.2**	**0.4**	**0.6**	**0.2**	**0.4**	**0.6**
CIFAR-10	CROSS-ENTROPY	83.88	72.09	57.17	88.76	81.7	69.05	77.89	56.32	39.80
	MW-NET	84.95	82.66	77.01	83.27	73.02	54.67	81.72	61.38	42.77
	L2B	88.83	86.27	80.23	91.23	90.51	89.23	89.46	85.79	74.23
	EMLC	81.60	71.24	58.55	88.51	80.95	68.11	80.32	74.09	61.57
	GLC	88.59	83.44	76.43	91.37	89.78	86.72	88.57	84.35	63.42
	FASTEN	86.19	77.62	70.04	91.63	88.82	85.71	86.82	81.16	61.88
	OURS(MIXED W/ τ)	91.09	86.06	81.49	93.01	90.37	57.76	89.84	74.17	46.00
	OURS(MIXED W/O τ)	89.96	84.80	78.58	92.25	86.31	68.75	90.97	73.15	40.95
	OURS(NEURAL)	**91.62**	**88.62**	**82.96**	**93.22**	**92.93**	**91.75**	**91.76**	**89.79**	**82.46**
CIFAR-100	CROSS-ENTROPY	62.34	46.40	28.16	63.04	48.05	29.94	58.75	40.57	22.00
	MW-NET	62.98	57.71	**47.76**	64.12	61.56	50.63	45.13	20.29	6.99
	L2B	64.28	54.38	20.66	64.66	58.11	43.33	**66.05**	**59.55**	**47.45**
	EMLC	62.80	46.51	31.74	62.94	47.09	32.34	61.94	49.34	28.44
	GLC	65.75	52.03	41.16	67.57	58.84	37.64	60.63	55.53	45.29
	FASTEN	59.37	43.65	26.09	61.91	43.27	31.50	59.70	45.79	33.15
	OURS(MIXED W/ τ)	**67.70**	**58.00**	45.36	**70.82**	**63.67**	53.23	63.39	42.65	19.31
	OURS(MIXED W/O τ)	65.89	56.62	39.73	67.53	59.88	43.65	59.24	39.29	18.68
	OURS(NEURAL)	64.89	57.26	47.26	64.61	61.73	**57.72**	64.19	57.23	46.42

Table 2. This table details the average GPU training time (in hours) for various methods using a single RTX 4090, with the relative ratio of time compared to our method.

METHOD	MW-NET	L2B	EMLC	GLC	FASTEN	OURS
TOTAL TRAINING TIME (RELATIVE TO OURS.)	1.08 (1.47X)	1.61 (2.20X)	1.09 (1.49X)	0.63 (0.87X)	**0.31 (0.43X)**	0.73

5.1.1. RESULTS

The overall results are presented in Table 1, and more details our methods are included in Appendix C.3. Additionally, we have quantified the training time for our method and compared it with existing methods. These results are presented in Table 2 and 9. Our approach secures a leading position in terms of training efficiency, necessitating less time compared to existing meta-learning-based methods. The bulk of the time consumption is attributed to the training of the noisy and clean posterior models. Notably, a relatively minimal amount of time is spent on optimizing the parameters in the outer loop, underscoring the method's computational efficiency.

The experimental outcomes distinctly illustrate the benefits of integrating a flexible temperature term within the label refinement process. For a more in-depth analysis, we provide visualizations of the weights and temperature changes during training in Figure 6 and 7. Additionally, our results clearly demonstrate that neural network correction significantly surpasses other methods in contexts with instance-dependent noise. Figure 5 illustrated a training dynamic, where our method is compared to MW-Net and

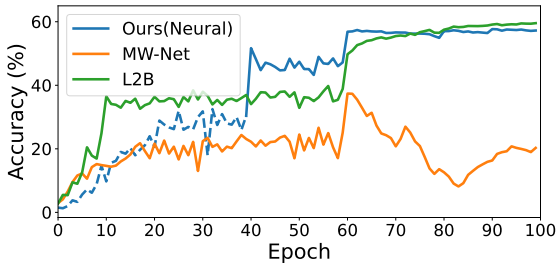

Figure 1. Comparison of training dynamics between our method, MW-Net and L2B on CIFAR-100 under instance-dependent noise with a level of 0.4. The dashed line represent the periord before loading the checkpoint in experiment on our method.

L2B, showcasing our approach's enhanced resistance to noise overfitting in such scenarios. The robustness of our method is largely attributable to the integration of diverse label sources, all effectively steered by the clean data. Considering the demonstrated effectiveness of our decoupled refinement framework, as evidenced by the statistical data in Table 9, it is practical for real-world applications to configure different forms of the function g to adapt to different noise categories. The optimal configuration can then be

Table 3. Comparative analysis of test accuracy on Clothing1M with state-of-the-art methods utilizing clean data. The symbol ✦ indicates the use of meta-knowledge, while ✓ denotes methods that directly train on clean instances with full forward and back-propagation applied to the classifier parameters.

METHOD	EXTRA DATA	ACCURACY (%)
CROSS-ENTROPY	✗	69.21
GLC	✓	74.26
FASTEN	✓	77.83
MW-NET	✦	73.72
FAMUS	✦	74.40
MLC	✦	75.78
DMLP-DIVIDEMIX	✦	78.23
EMLC	✦	79.35
OURS(NEURAL)	✦	**79.62**

selected based on the best performing metric on the clean dataset. This strategy enables a tailored approach to label correction, further enhancing the method's applicability.

5.2. Experiments on Clothing1M

Clothing1M (Xiao et al., 2015) is a real-world large-scale dataset where image instances come from online shopping websites where the noisy labels are devised by leveraging user tags as proxy annotations. To validate the efficacy of our proposed method in real-world, large-scale applications, we conducted extensive experiments detailed in this section. We use ResNet-50 pre-trained on ImageNet following previous works. Detailed configurations and settings employed during these experiments can be found in Appendix C. We opted for the neural network for the parameterization of g, driven by its enhanced capability to manage instance-dependent label noise which is common in real-world datasets. All the refinement optimization settings remain consistent with those used in previous experiments.

5.2.1. RESULTS

In addition to the baseline methods, we also consider other leading approaches (Xu et al., 2021; Zheng et al., 2021; Tu et al., 2023) for comparison. These results presented in Table 3. Particularly, the semi-supervised learning methods outshine conventional methods that rely on modeling label noise through transition matrices or weighting techniques. Figure 2 provides a detailed illustration of our model's training dynamics. A particularly notable finding is the substantial increase in model accuracy, exceeding 77% following the first training epoch utilizing the optimized $\hat{y}$. This rapid enhancement in performance underlines the efficacy of our approach. Furthermore, the figure reveals the high confidence levels in the output of g_θ, especially notable at an alpha value of 0.02. The central part of the figure effectively highlights the trajectory of the actual error of our model, demonstrating its close alignment with the evolving combined label space $\hat{y}$. These insights collectively emphasize the robustness and adaptability of our method, showcasing its capability to adeptly navigate through complex noisy labels in real-world scenarios.

Table 4. Performance variations based on different neural network configurations on Clothing1M. Here $\boldsymbol{R_M(\hat{y})}$ represents the empirical risk on the clean data.

METHOD	$\boldsymbol{R_M(\hat{y})}$	ACCURACY (%)
1 LAYER, $\{z_t\}$	0.5833	78.83
2 LAYERS, $\{z_t\}$	**0.5268**	**79.62**
2 LAYERS, $\{\widetilde{y}_t\}$	0.7519	75.24

Table 5. Performance variations based on different clean sample size on Clothing1M. Here $\boldsymbol{R_M(\hat{y})}$ represents the empirical risk on the clean data.

CLEAN NUMS	$\boldsymbol{R_M(\hat{y})}$	ACCURACY (%)
1400	0.2172	77.04
14K	0.4527	79.29
50K	0.5268	79.62

5.2.2. ABLATIONS

In our investigation of the refinement neural network's configurations, we focused on three distinct architectures: a single linear layer with softmax output, two linear layers with a non-linear activation function (used in our settings), and replacing input logits z_t with pseudo-predictions $\widetilde{y}_t$. The outcomes of these configurations are tabulated in Table 4. From these results, it becomes evident that the incorporation of nonlinear operations plays a crucial role in the refined performance. Additionally, it is noteworthy that z_t provides much more useful informative cues compared to the raw labels $\widetilde{y}_t$, implying that earlier layer outputs contain richer information than the final prediction.

Subsequently, we explored the impact of the clean data scale on our performance. This aspect was examined using two different sizes of clean sample sets: 1,400 and 14,000 samples. The corresponding results are detailed in Table 5. Interestingly, while a smaller empirical risk was observed on the clean dataset, an over-fit on this clean data led to a decrease in overall performance. This finding underscores the importance of balancing the influence of clean data within the training process to avoid performance degradation.

5.2.3. RESOURCE COMPARISON

In this section, we delve into a comparative analysis of Ours(Neural), L2B and EMLC (previous State-of-the-Art) regarding their training efficiency and memory usage on

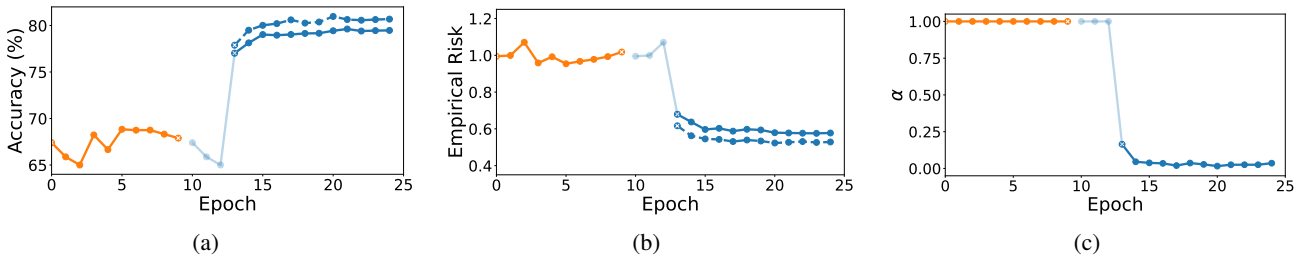

Figure 2. Training dynamics of our method on Clothing1M, depicted across phase 1 (orange line) and phase 2 (blue line). The semi-transparent areas denote the portions before the checkpoint. Figure (a) shows the accuracy (%) over epochs. Figure (b) presents the empirical risk on both the clean dataset and the test dataset. Figure (c) illustrates the evolution of the parameter α throughout training. Solid lines indicate metrics on test data, while dashed lines represent metrics on the limited clean data.

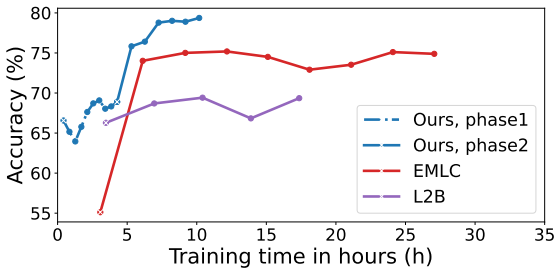

Figure 3. Training time comparison on the Clothing1M, showing the training durations for our method, L2B and EMLC. All periods of time consumption are counted for out method in this figure.

Table 6. Comparison of average memory usage for our method, L2B, and EMLC on Clothing1M under the specified training configurations.

METHOD	AVG MEMORY COST (MIB)
OURS	**16915**
L2B	24120
EMLC (*ICCV'23*)	22190 / 21813

Clothing1M. A key aspect of this comparison is the training configuration: our method and L2B utilize a single RTX 4090 GPU with 128 mini-batches, whereas EMLC employs a 2-GPU setup, with each GPU processing 64 mini-batches, and the other configurations remain the same with our settings and their original reports. This configuration ensures that the total batch size remains equivalent across both methods, providing a fair basis for comparison.

The training time comparison is presented in Figure 3, while Table 6 details the average memory usage for each method. Notably, other two MAML-based methods exhibited prolonged iteration times per epoch, while our framework demonstrated greater efficiency. The comparative results highlight our method's advantages in terms of performance, time efficiency and memory consumption, underscoring its potential for practical applications in large-scale settings.

Table 7. Test accuracy on Clothing1M with different compositions of the limited clean dataset. In this experiment, 50K samples are removed from the noisy training dataset.

CLEAN DATA COMPOSITION	ACCURACY (%)
50K CLEAN	79.42
45K CLEAN + 5K NOISY	79.33
40K CLEAN + 10K NOISY	78.59
30K CLEAN + 20K NOISY	77.50

5.2.4. CORRUPTION LEAK ON THE CLEAN DATASET

In this section, we explore the impact of corrupting the clean instances on the performance of our methods. Specifically, we extract 50k noisy instances out of the 1 million instances in the noisy training dataset, ensuring they are not used for training. These noisy instances are then used to replace the clean instances at varying corruption rates: 0%, 10%, 20%, and 40%, in order to make a comprehensive evaluation.

The results of these experiments, presented in Table 7, illustrate the impact of different levels of clean data corruption on the performance of our method. Notably, the findings indicate that our approach maintains strong performance in the presence of mild corruption in the clean dataset. This outcome highlights the robustness of our method, affirming its efficacy in practical scenarios.

6. Conclusion

Our research introduces an innovative approach for refining noisy labels in noisy label learning, which diverges significantly from conventional methodologies on the use of clean data. Central to our contribution is the development of a self-training framework that incorporates decoupled bi-level optimization with novel utilization on limited clean data to guide the refinement process. Our experiments validate the effectiveness and efficiency of our method. This approach not only improves accuracy in noisy label scenarios but also reduces computational overhead, offering a robust solution for large-scale applications.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Two General Assumptions on Types of Label Noise

Instance-Independent Noise: Under this assumption, the observed noisy label $\widetilde{y}$ is conditionally independent of the feature vector x, such that $p(\widetilde{y}|y,x) = p(\widetilde{y}|y)$. This implies that the probability of encountering a noisy label $\widetilde{y}$ for a given true label y is consistent across all feature vectors, suggesting a uniform distribution of label noise irrespective of individual data instances.

Instance-Dependent Noise: This scenario considers the probability of label noise to be influenced by the specific features of each instance, with the noise probability $p(\widetilde{y}|y,x)$ varies with the feature vector x. This scenario acknowledges that label noise can be affected by factors like annotator biases or automated labeling errors which is more reflective of real-world conditions

B. Complete Explanation of Current Methods Utilizing Limited Clean Data

B.1. Meta-learning-based Label Correction Methods

Meta-learning approaches typically optimize hyper-parameters using a clean dataset as a form of "reward" data. This process involves learning a meta-model that can adaptively adjust the learning process based on the feedback from the clean dataset. In the domain of the widely designed strategies, label correction techniques emerge as a particularly distinguished subset. These methods have consistently demonstrated superior performance, distinguishing themselves by effectively leveraging clean datasets to enhance label accuracy (Zhang et al., 2020; Taraday & Baskin, 2023). The fundamental mathematical representation of these methods can be encapsulated as:

$$\Theta^* = \arg\min_{\Theta} \frac{1}{M} \sum_{i=1}^{M} \mathcal{L}(f^*_{\Theta}(x_i^{(m)}), y_i^{(m)}), \tag{13}$$

$$f^*_{\Theta} = \arg\min_{f \in \mathcal{F}} \frac{1}{N} \sum_{i=1}^{N} \ell(f(x_i), \hat{y}_k(\Theta)). \tag{14}$$

Here, Θ represents the set of hyper-parameters being optimized. Although the meta-learning-based methods are powerful in leveraging clean data, they often incur intensive computation and high memory costs due to high-order gradient computations and multiple back-propagations in the learning process. This aspect makes them less favorable for large-scale datasets. Although some recent studies (Xu et al., 2021) have attempted to mitigate these issue, the fundamental challenge of high computational overhead remains.

B.2. Transition-matrix-based Correction Methods

These methods, which focus on correcting noisy labels using a transition matrix $T \in [0,1]^{c \times c}$ (Van Rooyen & Williamson, 2017) , represent a significant approach in the domain of label noise management. The transition matrix serves to bridge the class posterior probabilities for noisy and clean data, as shown in

$$p(\widetilde{y}|x) = T^\top p(y|x). \tag{15}$$

The core of this method is encapsulated in the following mathematical formulation:

$$\begin{aligned} T_{(k,j)} &= p(\widetilde{y} = j | y = k) \\ &\approx \frac{1}{|D_k^{(m)}|} \sum_{x \in D_k^{(m)}} p(\widetilde{y} = j | x), \end{aligned} \tag{16}$$

$$f^*_T = \arg\min_{f \in \mathcal{F}} \frac{1}{N} \sum_{i=1}^{N} \ell(T^\top f(x_i), \widetilde{y}_i), \tag{17}$$

where $D_k^{(m)}$ represents the sub-dataset of class k. However, these methods face several challenges. Firstly, it primarily addresses instance-independent noise, which limits its applicability in scenarios where noise is dependent on specific instances (Yao et al., 2021). Moreover, even when only considering an instance-independent scenario, although it ensures the *classifier-consistent* property (Xia et al., 2019), transition-based methods are significantly impacted by the variance inherent in the one-hot noisy labels, leading to sub-optimal performances in practical applications, as shown in Section. 5.1.1.

C. Experiment Details

C.1. Implementation Details and Hyper-parameters

We detail the specific implementation aspects and hyper-parameters used in our experiments across various datasets. Table 8 outlines the key training parameters and settings employed for CIFAR-10, CIFAR-100, and Clothing1M datasets.

Table 8. Experimental setup and hyper-parameters for training. The setting of batch size on Clothing1M is changed to 128 for the experiments in Section. 5.2.3, while other hyper-parameters are the same.

DATASET	CIFAR-10	CIFAR-100	CLOTHING1M
ARCHITECTURE	RESNET-34	RESNET-34	RESNET-50
PRETRAINING	NONE	NONE	IMAGENET
PHASE1 EPOCHS	100	100	10
PHASE2 EPOCHS	100	100	15
(START FROM PHASE1 CHECKPOINTS)	(40)	(40)	(3)
POOL UPDATE FREQUENCY	10	10	1
TRAINING AUGMENTATIONS	HORIZONTAL FLIP RANDOM CROP	HORIZONTAL FLIP RANDOM CROP	RESIZED CROP HORIZONTAL FLIP RANDOM COLORJITTER RANDOM GRAYSCALE
OPTIMIZER	SGD	SGD	SGD
BATCH SIZE	128	128	256
MOMENTUM	0.9	0.9	0.9
WEIGHT DECAY	$5e^{-4}$	$5e^{-4}$	$5e^{-4}$
INITIAL LR	0.1	0.1	0.01
SCHEDULER	LR-STEP	LR-STEP	LR-STEP
SCHEDULER MILESTONE	$60, 80$	$60, 80$	$5, 10$
SCHEDULER STEP-DECAY	0.1	0.1	0.1

C.2. Time Consumption Details

For our experiments on CIFAR and Clothing1M, all models were trained using a single RTX 4090 GPU. We provide a detailed breakdown of the time consumption for different stages of our training process in our experiments on CIFAR-100 and Clothing1M in Table 9 and Table 10. The comprehensive time distribution is essential for evaluating the overall computational demands of the algorithm, as outlined in Algorithm 1.

Table 9. Detailed comparison of training time on the CIFAR-100 dataset. This table presents the average GPU computation time (in minutes) of different stages. The training time on GPU and the proportion of each stage are also provided for a holistic view of the time allocation.

PERIOD	TRAINING h	TRAINING f	INFERENCE ON $\widetilde{D}$ AND $D^{(m)}$	OPTIMIZING α^*, g^*	TOTAL
TIME CONSUMPTION (PERCENTAGE)	18.8 43.0%	20.0 45.6%	0.17×6 0.02%	0.67×6 0.09%	43.8

Table 10. Detailed comparison of training time on Clothing1M. The table enumerates the all the time spent (in hours) on various stages. The total training time and the proportion of each stage are also provided for a holistic view of the time allocation.

PERIOD	TRAINING h	TRAINING f	INFERENCE ON $\widetilde{D}$ AND $D^{(m)}$	OPTIMIZING α^*, g^*	TOTAL
TIME CONSUMPTION (PERCENTAGE)	4.51 29.4%	6.67 44.3%	0.16×12 12.8%	0.17×12 13.6%	15.05

C.3. Additional Results

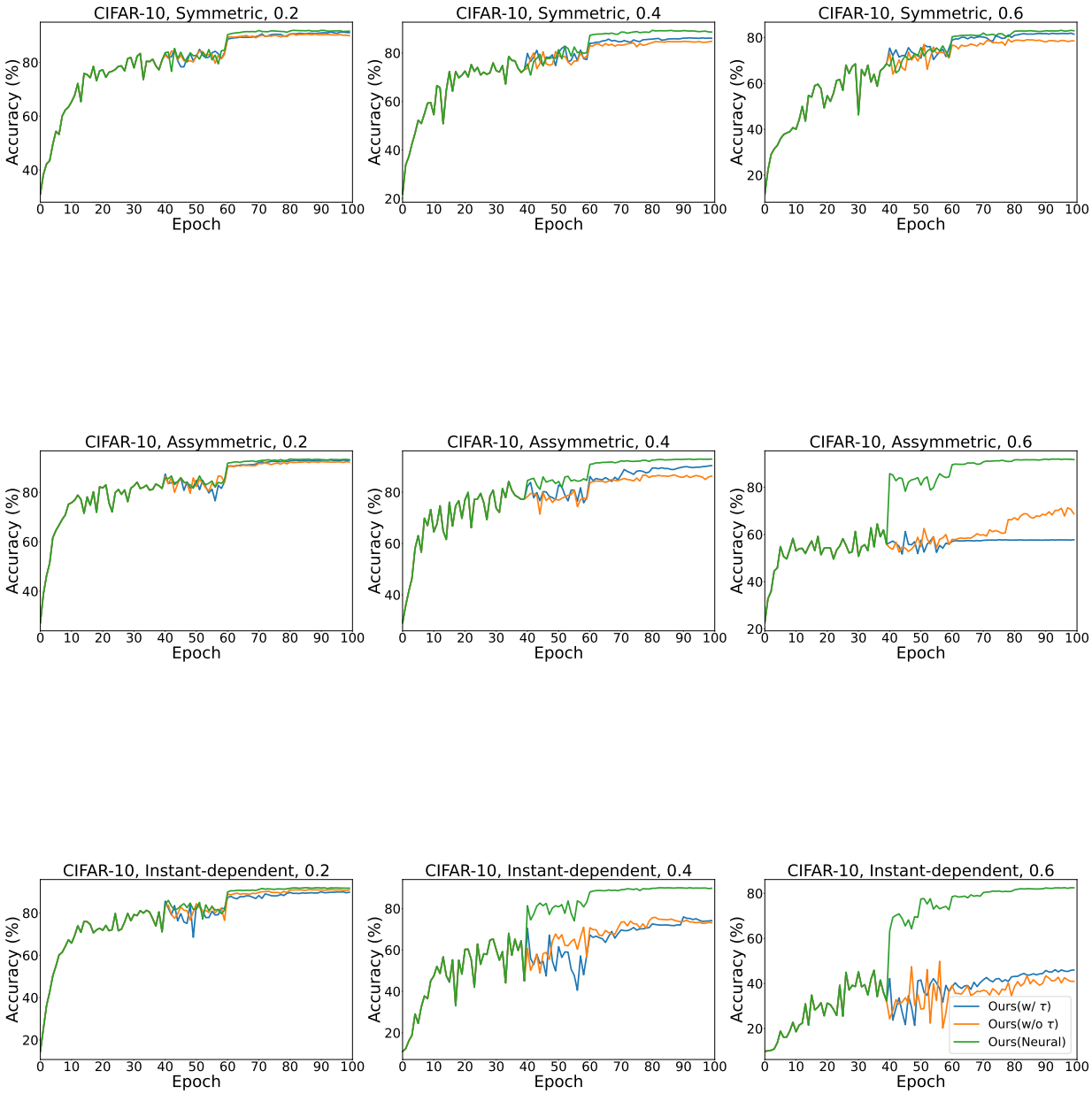

Figure 4. Training dynamics of our methods on CIFAR-10 under different noise scenarios.

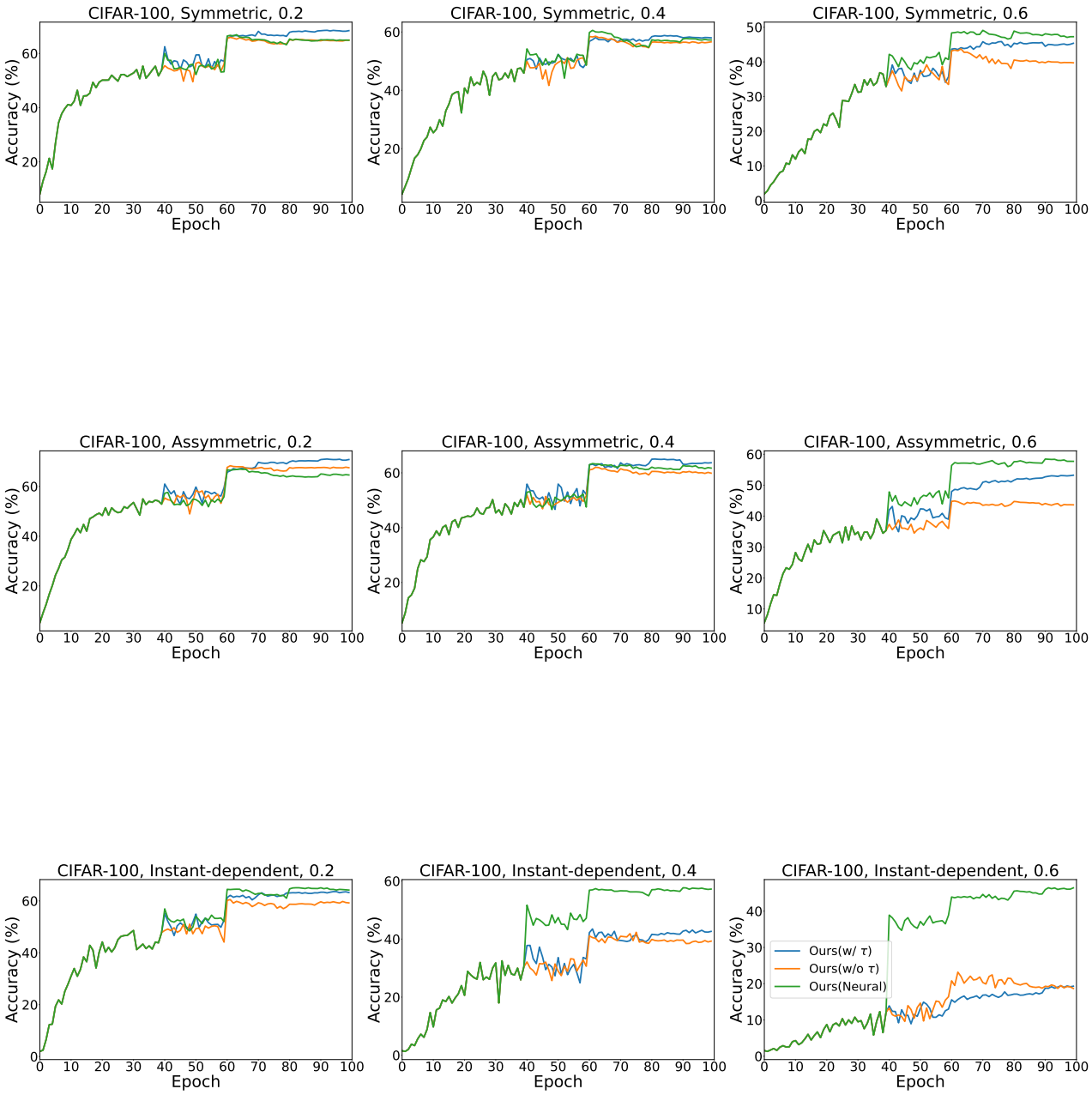

Figure 5. Training dynamics of our methods on CIFAR-100 under different noise scenarios.

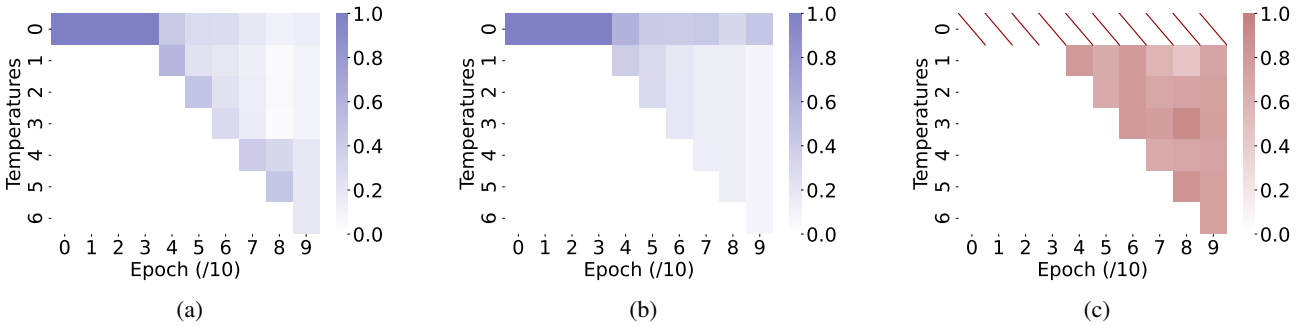

Figure 6. Visualization of weights and temperatures in the mixed-up strategy, both with and without adjustable τ, applied to CIFAR-100 under symmetric noise level of 0.4. Row 0 represents the weights assigned to the original noisy label. Figures (a) and (b) depict the evolution of weights in the strategies with and without τ, respectively. Figure (c) illustrates the changes in τ over time.

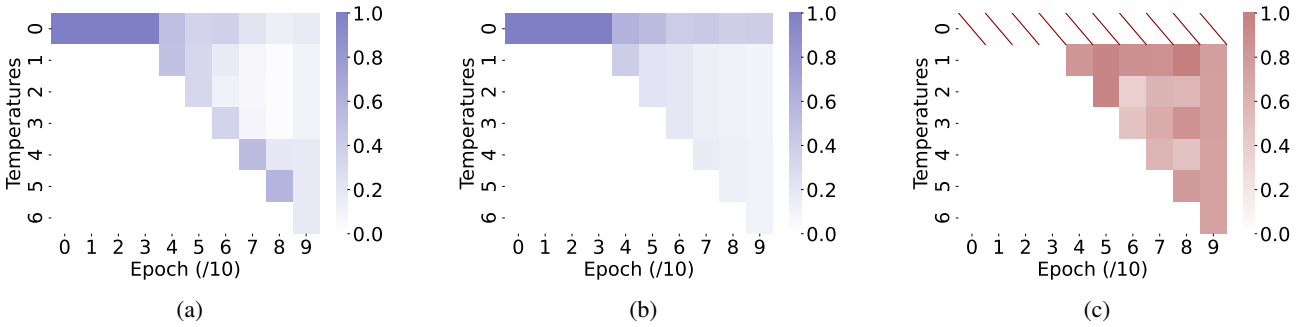

Figure 7. Visualization of weights and temperatures in the mixed-up strategy, both with and without adjustable τ, applied to CIFAR-100 under asymmetric noise level of 0.4. Row 0 represents the weights assigned to the original noisy label. Figures (a) and (b) depict the evolution of weights in the strategies with and without τ, respectively. Figure (c) illustrates the changes in τ over time.